# Online Supportive Interactions: Using a Network Approach to Examine Communication Patterns Within a Psychosis Social Support Group in Taiwan

**Hui-Jung Chang**
*Department of Journalism and Communication, Fu-Jen Catholic University, Taipei, 24205 Taiwan.*
*E-mail: hj.chang.taipei@gmail.com*

**A network approach was used to determine the overall supportive communication patterns constructed within the PTT psychosis support group in Taiwan, the largest bulletin board system in the Chinese-speaking world. The full sequences of supportive interactions were observed over a $2\frac{1}{2}$-year period from February 2004 to July 2006. The results indicated that the most exchanged support types were information and network links. All types of supportive communication networks were relatively sparse, yet small groups of cliques with different provision of support types formed within the psychosis group. Most of the online supportive interactions exchanged at dyadic and triadic levels. The overall supportive network was highly centralized. The overall findings with implications for future studies were discussed.**

## Introduction

Humans need various types of social support at many points in life, especially when they or the ones they care for are in need of health information (Yoo & Robbins, 2008). Individuals purposively acquire health information from "selected information carriers" (Johnson, 1996, p. 9) to reduce uncertainty regarding health issues. Indeed, much has been written about the associations between social support and individuals' emotional and physical well-being (Burleson & MacGeorge, 2002).

In general, social support is important across age (Tichon 1& Shapiro, 2003; Wright, 2000; Xie, 2008), and culture (Burleson, Liu, Liu, & Mortenson, 2006; Hiller & Franz, 2004), and is especially significant to individuals afflicted with diseases (Bambina, 2005; Bertero, 2000; Eichhorn, 2008). "Indeed, a life without any supportive relationships is virtually unthinkable" (Albrecht, Burleson, & Goldsmith, 1994, p. 419). Fortunately, with the advent of the Internet, which has "rapidly expanded to address the demand for medical information on health-related topics" (Cotton & Gupta, 2004, p. 1796), individuals have access to a superior and fabulous medium for social support (Robinson & Turner, 2003; Walther & Parks, 2002).

Online social support has fundamentally changed two structural aspects of traditional face-to-face (FtF) social support: the channels of communication and the sociometric relationships of support seekers and providers (Walther & Parks, 2002). These two features both expand individuals' personal social support networks. First, individuals can seek and provide social support without constraints of time and space in online venues (Eichhorn, 2008; Franzen, 2000; Wellman, 2001). Second, individuals can have more social choices regarding to whom they could go when they engage in supportive interactions (Hlebec, Manfreda, & Vehovar, 2006).

The other change brought about by online supportive interactions, from a researcher's point of view, is that the full sequence of events comprising supportive interactions can be easily recorded and observed online. In general, researchers have agreed that there are four sequences of events for supportive interactions: support activation target, support provision by a helper, support receipt and accompanying reactions by the target, and responses to the target's reaction by the helper. However, in the past, most studies have focused on the first two phases of the full sequence, and few studies have been conducted to study the full four phases of supportive interactions (Burleson & MacGeorge, 2002). One major reason for the paucity of research in this area is probably the challenge of observing and recording full sequences of naturally occurring supportive interactions between people in real life.

Yet, "the existence of electronic groups provides an opportunity unobtrusively to observe the dynamics of particular support groups" (Galegher, Sproull, & Kiesler, 1998, p. 494), especially when most dedicated online support groups exchange social support via a series of "bulletin board-type

Received June 26, 2008; revised February 12, 2009; accepted February 16, 2009

© 2009 ASIS&T • Published online 13 April 2009 in Wiley InterScience (www.interscience.wiley.com). DOI: 10.1002/asi.21070

arrangement of messages" (Walther & Boyd, 2002, p. 155). On electronic bulletin boards, users post messages for others to read and reply to. When a post initiates chains of replies discussing the same topic, threads of communication are formed. In other words, in the context of social support groups, a communication network of supportive interactions is constructed through posts exchanged via questions or remarks and replies or responses. Treating Usenet discussion groups as networks, Himelboim (2008) described Usenet participants "in terms of their place in a network, the web of postings and replies" (p. 159).

Thus, in this study, a network approach was used to determine overall online supportive-communication patterns. The basis of social network analysis is who speaks to whom, about what, using which communication mode. Accordingly, this study attempted to employ both social network analysis and content analysis, with the former emphasizing the detection and interpretation of patterns of social linkages among individuals (Haythornthwaite, 2000; Nooy, Mrvar, & Batagelj, 2005) and the latter emphasizing communication content (Bauer, 2000; Herring, 2004), to determine who posts what type of social support to whom within an online support group. For a more fine-grained analysis, one or more types of social relations exchanged by the same set of individuals also were examined (Monge & Contractor, 2003). Six networks in total were analyzed: a multiplex network, which contains all types of social support, and five uniplex networks for each type of social support.

In Taiwan, statistical data reported by the Department of Health has indicated an average increase of 10.8% for individuals hospitalized for psychosis-related syndromes in the past 5 years (from 2001–2005) (www.doh.gov.tw). For this reason, the online support group selected for examination was the psychosis group located at PTT, the most popular Bulletin Board System (BBS) in Taiwan and the largest BBS in the Chinese-speaking community. In addition, Taiwan is a country characterized by a collectivistic culture and high-context communication (Hall, 1976; Hofstede, 2001). In high-context communication, individuals communicate in a relatively indirect, implicit, and ambiguous way. How do the Taiwanese seek and provide social support online when they are afflicted with such a sensitive issue as psychosis? Thus, the full sequences of supportive interactions within the PTT psychosis group were observed over $2\frac{1}{2}$-year period, from February 2004 to July 2006. Specifically, four communication-network measures related to traditional social support (Wellman, 1992), core discussion networks (Marsden, 1987; McPherson, Smith-Lovin, & Brashears, 2006) and information flow (Borgatti, 2005) were used: size, density/connectedness, clique, and centralization.

In the next section, I first discuss the associations between supportive interactions and communication networks. Next, I introduce types of social support and then present the network measures utilized to describe the PTT psychosis network. Research questions will be proposed accordingly at the discussion of each network measure.

## Literature Review

### *Supportive Interactions and Communication Networks*

The concept of social support originated in the 19th century, when English and French demographers implied the notion of communication links in proclaiming that "marriage is a health-promoting institution" (Burleson & MacGeorge, 2002, p. 376). Durkheim (1897), in his classic thesis *Suicide*, noticed the associations between individuals' social networks and their health status. He suggested that an individual's social links would gradually decrease in an industrialized society. As a result, individuals would tend to commit suicide when they did not have anyone with whom to communicate in their social networks.

The notion of communication networks has been interpreted more explicitly in studies of social support since the mid-1970s, when pioneering researchers in sociology, psychology, and communications deliberated over the definition of social support (Burleson & MacGeorge, 2002). Among them, Moss' (1973) work on health and social integration stated that individuals mostly learn values and norms from the social communication networks in which they participate. Cobb (1976) defined social support as "information leading the subject to believe that he is cared for" (p. 300).

Caplan's (1974) work on support systems and community mental health has suggested that communication networks should be built among educators and students so that students are prepared for unexpected problems in life. Furthermore, viewing support as a communication phenomenon, Albrecht and Adelman (1987) defined social support as "verbal and nonverbal communication between recipients and providers" (p. 19). Different types of social support have been argued to meet different kinds of needs throughout life.

*Types of social support.* Cutrona and Russell (1990) and Cutrona and Suhr (1992) observed different types of social support provided to individuals who had just lost their jobs or family members and proposed an "optimal matching model of stress and social support" where "different kinds of supportive messages would be maximally beneficial following different kinds of stressful events" (Cutrona & Suhr, 1992, p. 155). Following an intensive review of the relevant literature (e.g., Cobb, 1976), Cutrona and Suhr listed five different types of social support mostly used in supportive interactions: (a) informational support, comprising suggestions, advice, and factual inputs; (b) tangible support, such as the provision of necessary goods (e.g., money, food, books); (c) esteem support, which includes validating support seekers' worth by exerting positive communication behaviors; (d) network support, involving the expression of companionship, connection, and belonging; and (e) emotional support, comprising the expression of affection, sympathy, listening, understanding, empathy, and encouragement.

Cutrona and Suhr (1992) also suggested that these five types of social support can be further classified into two broader categories: action-facilitating support and nurturant support. Action-facilitating support aims to solve or eliminate


DOI: 10.1002/asi

problems facing support seekers; informational and tangible support are in this category. Nurturant support aims to comfort or console support seekers; emotional, network, and esteem support are included in this category.

This typology of social support has received empirical verification within online support groups, especially in relation to groups of individuals afflicted by various kinds of health problems. For example, Coulson (2005) found that within a computer-mediated support group for individuals living with irritable bowel syndrome, informational support was most frequently provided. Informational support also was more prevalent for individuals suffering from an eating disorder (Eichhorn, 2008). In addition to informational support, emotional support was frequently transmitted within an online social support group for people with physical disabilities (Braithwaite, Waldron, & Finn, 1999). Other than informational and emotional support, Bambina (2005) found that companionship was a major type of support among cancer patients. Moreover, Bjornsdottir (1999) reported that appraisal support, which includes affirmation and feedback, was the major type of support received among individuals concerned about heart disease.

### *Properties of Communication Networks*

Communication networks are "the patterns of contact that are created by the flow of messages among communicators through time and space" (Monge & Contractor, 2003, p. 3). Thus, the pattern of contacts created by informational messages among social support group members forms an informational communication network; likewise, the pattern of contacts created by emotional messages creates an emotional communication network. These communication contacts, developed by individuals within a social system, usually influence and determine the communication behaviors in the system. Yet, to date, in addition to the overall distribution of types of social support within an online support group, we have limited understanding of online supportive-interaction networks.

Four network indicators were used for the present study: size, density/connectedness, clique, and network centralization.

*Size.* Network size traditionally has been used to describe the range of individuals' communication networks, within which they discuss important matters (Marsden, 1987). The first study to examine different aspects of the core discussion networks of Americans was conducted in 1985, utilizing General Social Survey network data (Marsden, 1987). Three network measures were used to describe the core discussion networks: size, density, and composition. Network size provides a direct measure of social integration and correlates with the availability of social support at the individual level (Network density and composition will be discussed alongside other network indicators used in the present study.)

The 1985 results indicated that nearly half (49.1%) the individuals surveyed discussed important matters with less than three people. Nearly two decades later in 2004, researchers found that an even larger percentage (62.8%) of individuals had discussion networks consisting of less than three people (McPherson et al., 2006). While the American population has lost discussion partners over time, what has happened to the size of Taiwanese online psychosis networks?

In a study to examine the social support networks in Slovenia, Hlebec et al. (2006) reported that on average, Internet users had 6.7 people in their social support networks, slightly higher than 6.3 people for non-Internet users. Compared to the aforementioned numbers, a larger size of online support networks ranging from 30 to 733 has been reported in previous studies of bulletin board type support groups (Table 1). First, Bjornsdottir (1999) observed support messages posted in a 4-week period within a heart-disease support group on a commercial health Web site in the United States. She gathered a sample of 69 messages posted by 30 individuals in the 4 weeks of the study. Within a similar sampling interval, Braithwaite et al. (1999) reported 1,472 messages posted by 42 individuals in a support group for people with disabilities during a 1-month period in the United States. In addition, Coulson (2005) collected a sample of 572 messages written by 132 individuals in an 8-month study period posted by individuals affected by irritable bowel syndrome. These 132 individuals were from several European and North American countries. In addition, Eichhorn (2008) analyzed the strategies used to solicit social support and the type of social support provided on the top five Yahoo! eating-disorder groups over time, using a random sample of 490 postings. Finally, Galegher et al. (1998) observed three Usenet support groups for a 3-week period in the United States, and the total messages and participants for the three groups were 200 messages and 119 individuals for an arthritis group, 5,520 messages and 274 individuals for an attention deficit group, and 39,864 messages and 733 individuals for a depression group.

As a basis for comparison, this study researched the size of Taiwanese online psychosis networks. Thus, the first research question is:

> **RQ1:** What is the size of the multiplex PTT psychosis network and each of the five uniplex networks (informational, emotional, esteem, network, and tangible), and how many messages were exchanged within each?

In addition, the PTT psychosis network was composed of multiple threads. A descriptive analysis will be performed on these threads so that a more detailed examination of supportive interactions can be provided. Accordingly,

> **RQ2:** What are the sizes of these threads? How many individuals and messages are involved within the shortest/longest threads?

*Density/connectedness.* Network density is another measure used to describe the concentration of individuals' interpersonal networks. It also relates directly to well-being and the availability of social support (Marsden, 1987). Network


DOI: 10.1002/asi

TABLE 1. Previous bulletin board-type studies: Network measures.

| Studies | Support group | Sampling period | Size | No. of messages | Density[a]/connectedness |
|---|---|---|---|---|---|
| Bjornsdottir, 1999 | heart disease | 4 weeks | 30 | 69 | .08 |
| Braithwaite et al., 1999 | disabilities | 1 month | 42 | 1,472 | .85 |
| Coulson, 2005 | irritable bowel syndrome | 8 months | 132 | 572 | .03 |
| Eichhorn, 2008 | eating disorder | Longitudinal | N/A | 490 | N/A |
| Galegher et al., 1998 | arthritis | 3 weeks | 119 | 200 | .01 |
| | attention deficit | 3 weeks | 274 | 5,520 | .07 |
| | depression | 3 weeks | 733 | 39,864 | .07 |

[a]densities were roughly estimated by this author; they were not provided in the original articles.

density is defined as the mean intensity of linkage strength among discussion partners (McPherson et al., 2006). More linkages between individuals creates a tighter and dense structure (Nooy et al., 2005). In turn, the intensity of this tight and dense structure affects "normative pressures—both in terms of pressure to conform and the responsibility for support in times of need" (McPherson et al., 2006, p. 357). The mean density of the core discussion networks in 1985 and 2004 was .6 and .66, respectively, with values ranging from 0 (*a sparse network*) to 1 (*a high-density network*). Compared to the densities reported in the five online support networks reviewed earlier (see Table 1; densities range between .01 and .08), the FtF support networks were densely interconnected.

It has been suggested in the traditional social support literature that the effects of density are dependent on the types of social support provided. Thus, for example, delivery of chronic healthcare services was provided more in high-density networks while emotional aid was forthcoming in low-density networks. The argument is that greater service tended to be more widely provided in high-density networks, and more companionship was provided in low-density networks (Wellman, 1992). Few studies have examined the association between network density and types of social support within online support networks. However, if we follow the argument put forward in the traditional support literature, will the density of more informational-oriented networks (i.e., informational, tangible) be higher than the density of more companionship-oriented networks (i.e., emotional, esteem, network)? Thus, I propose the following set of research questions:

**RQ3.1:** What is the density of the multiplex PTT psychosis network and each of the five uniplex networks?

**RQ3.2:** Are there associations between density and different types of social support networks?

*Clique.* While density describes the number of linkages in a network, the concept of a clique captures the densest sections in a network. According to Hanneman and Riddle (2005), a clique is "a sub-set of a network in which the actors are more closely and intensely tied to one another than they are to other members of the network", and a subgroup of individuals in the network who form dense links among themselves. Thus, a clique is an indicator of network cohesion, serving as an explanatory variable for studying the emergence of consensus among participants within a group (Wasserman & Faust, 1994). It is expected that the more cohesively individuals are tied into a network, the more homogenous they are in developing attitudes and beliefs. Therefore, clique analysis aids in the investigation of the social forces that drive some participants to "stick together." In traditional FtF social support, the composition of cliques could be kin, friends, and neighbors (Wellman, 1992).

Two general types of social relations, kin and nonkin, were examined in the 1985 and 2004 core discussion network studies (Marsden, 1987; McPherson et al., 2006). The kin networks were the most heavily interconnected over time. Individuals tended to discuss important matters with their spouse, parent, sibling, child, and other family members. The density of nonkin increased from 1985 to 2004, indicating that "the network's non-family discussion partners are now more likely to know each other and be close" (McPherson et al., 2006, p. 361). However, when most online support groups are so-called *weak-tie networks*, which are groups of loosely connected strangers whose "only connection is their common affliction or concern over a source of personal discomfort" (Walther & Boyd, 2002, p. 154), will cliques emerge in the PTT psychosis network? If they do emerge, is it because clique members shared psychosis affliction or because they built consensus in terms of providing the same types of social support during the interaction process? Thus, instead of examining clique composition, I ask the following research questions:

**RQ4:** Do cliques form in the multiplex PTT psychosis network and in each of the five uniplex support networks? If they do, what are the sizes of the cliques, and what type of support communication was exchanged within the cliques?

*Network centralization.* Supportive communication messages flow through the PTT psychosis network via postings and replies. To further understand the communication flow of the PTT psychosis networks, network centralization was examined. This is an "umbrella concept that examines the variation in individuals' centralities within a network" (Monge & Contractor, 2003, p. 44). A centralized network means that a few individuals have relatively higher centrality scores than do others in a network. Individuals with a high


DOI: 10.1002/asi

degree of centrality are the "stars" of their networks, and are often the most important or influential due to the social benefits resulting from their advantageous positions of information flow (Wasserman & Faust, 1994). For example, filling structural holes (Burt, 1992) enables individuals to directly link to two or more unconnected others. This position enables them "to control or, at least, influence the flow of information and knowledge among these others" (Monge & Contractor, 2003, p. 143).

In a traditional interpersonal environment, individuals rely more on kinship as a source of influence to discuss important matters. What happens in an online weak-tie environment where groups of loosely connected strangers seek and provide social support? Do the supportive messages flow through the PTT psychosis network through its "stars?" Previous studies have explored the functions and emergence of these stars. In her study of an Internet cancer discussion forum, Bambina (2005) found that one star emerged and played a key role linking together a highly bifurcated network. Additionally, in a study to discern the emerging health information-seeking network among 24 "stay-at-home moms" (Tardy & Hale, 1998), opinion leaders emerged who served not only as the information center (e.g., regarding baby care) but also the liaison linking two cliques.

With regard to the PTT psychosis board, the following research question is proposed:

> **RQ5:** Do stars emerge in the multiplex PTT psychosis network and in each of the five uniplex support networks? Do they have an impact on the overall supportive interactions?

## Method

### *Research Site*

Founded in 1995 in Taiwan, PTT.CC (telnet://ptt.cc/) is the largest BBS in Taiwan as well as in the Chinese-speaking community. It hosts more than 30,000 discussion boards and has more than 600,000 registered users. On average, more than 1,000,000 individuals visit the site every day. The mean age of PTT users is 21 years.

The psychosis discussion group, set up in PTT in 2003, aims to provide a discussion space for people suffering emotional distress as well as an alternative channel to acquire updated and accurate psychology-related information.

### *Data Collection and Sampling*

Since Computer-Mediated Communication (CMC) is "irresistible in its availability" (Jones, 1999, p. 13) and to maintain continuity in relationships between and among the psychosis participants (Johnson, 1993), the present study examined all posts that remained on the discussion board when the study started in August 2006. As a result, the sampling period was from February 2004 to July 2006. To meet the goal of this study, posts that addressed problems or sought support on behalf of other people (e.g., relatives, friends) were excluded from the analysis. Posts that contained advertisements for medications, doctors, and hospitals and public announcements irrelevant to the aim of the board also were excluded. A total of 689 posts written by 438 individuals were collected. To ensure the confidentiality of the PTT psychosis users, all names used in the posts were replaced with numbers during the analysis process. Each individual was assigned a number based on the time order in which the post appeared on the board during the sampling period. These posts were then classified into two groups: threads and isolates. Threads are posts that consist of chains of valid replies, automatically sorted by the PTT computer systems. The shortest thread consists of two posts: one initiating the support-seeking process and one reply. Isolates are posts that ask for support, but receive no replies. Among the 689 posts, 168 threads consisting of 558 posts and 131 isolate posts were found. The 168 threads were written by 344 individuals, and 131 isolates were posted by 113 individuals. The 168 threads involving supportive interactions among 344 individuals were retained for analysis.

### *Content Analysis and Types of Social Support*

*Category construction.* The five types of social support defined by Cutrona and Suhr (1992), and Cutrona and Russell (1990) were used to categorize communication content within the PTT psychosis group. After preliminary examination of the posts on the board, the tangible support category was deleted because it was not used during the sampling period. Then, due to the relatively large amount of posts containing a gesture of gratitude, the category of *thanks* was added. As a result, five types of social support were used in the final analysis, as follows:

**Informational support.** Posts provide information on medication, physicians/psychologists, and health institutions. For example,

> Subject Re: Questions on DSM-four???
> Time: Mon Sep 26 2005
> . . . It does have the effect of making people feel better. As for the side effect of vomiting, it may be due to your personal resistance to this medication. . . .

**Esteem support.** Posts provide positive comments to praise the support seeker's abilities or to alleviate their feelings of guilt. For example,

> Subject: Re: Mom says that I am a psycho
> Time: Mon May 2 2005
> . . . You have to learn to love yourself. Don't blame yourself for everything. Appreciate who you are and what you own in your life. You must have good qualities in yourself. You have a bright future ahead of you. The fact that your parents' marriage did not work out, does not mean you will not have a happy marriage yourself. Be strong. Life is still beautiful no matter how old you are. . . .

**Network support (access, presence, companions).** Posts provide messages to broaden the support seeker's social


DOI: 10.1002/asi

network so that they do not feel they are alone in the world. For example,

> Subject: Re: I have panic disorder
> Time: Sun Jul 31 2005
> . . . You can Google on the net to find lots of similar cases. Actually, when you start the medication, you will feel much better. One of my friends had this disease a while ago, but he is much better after the medication process. So, don't worry. Go! Go!

**Emotional support.** Posts provide understanding of a situation, express sorrow for the support seeker's situation, or provide them with hope and confidence. For example,

> Subject: Re: Am I sick?
> Time: Sun Jul 9 2006
> I understand what you have been through. My condition was much worse than you . . . I did not know what to do at that time . . . But after starting the medication process and exercises every day . . . Although I still feel that I am a loser, I survive. I wish you a better life.

**Thanks.** Posts express gratitude. For example,

> Subject: Re: I feel I am not a normal person
> Time: Thu Jun 22 2006
> Thank you very much for all of your answers. I have been thinking about my condition all day today, and conclude that I may have some personality disorder problem.

*Analysis*

Two graduate research assistants were in charge of the coding process. They were instructed to use Cutrona and Suhr's (1992, p. 161) social support category system and the category of thanks to code the 558 posts contained in the 168 threads. Hosti's (1969) formula, which accounts for agreement between two coders, was used to compute intercoder reliabilities. Satisfactory intercoder reliabilities were obtained for the categories: informational (.86), esteem support (.89), network support (.96), emotional (.89), and thanks (1.0).

*Network Analysis and Communication Network Measures*

*Measurement.* An adjacency matrix containing the same number of rows and columns was created month by month for each uniplex network. The cells of the matrix constituted the supportive links between two individuals, with the content coded by social support type (described later). The links were established when a post initiated a thread. The thread initiator was the message sender, and those who replied to the initiator's post were receivers. Supportive messages flowed through the PTT group by parallel duplication (Borgatti, 2005). That is, the message sender was assumed to send a message to all of the thread members in parallel, like a radio broadcast. When a thread member replied to the initiator, a reciprocal link was established between the thread member and the initiator.

The month-by-month matrix then was added across the sampling period (30 months), based on different types of social support, to generate the final five types of uniplex networks. Finally, the multiplex network, containing the five types of networks, was generated by adding together all of the final five uniplex networks. Four indicators (size, density/connectedness, clique, and network centralization) were used to analyze the PTT psychosis network, as follows.

**Size.** Size refers to the number of individuals involved in exchange linkages in a network.

**Density/connectedness.** This is the ratio of the number of actual communication links to the number of possible communication links in the network. The formula for density is $2L/g(g-1)$, where $L$ is the number of actual communication links and $g$ is the number of individuals in a network (Wasserman & Faust, 1994).

**Clique.** This is the largest connected subset of individuals and communication links. A clique contains at least three individuals, such that "mutual dyads are not considered to be cliques" (Wasserman & Faust, 1994, p. 254). All individuals in a clique are connected, and no individuals have communication links to individuals outside the clique. A "maximal complete subgraph" algorithm was employed by the UCINET program to determine cliques, in which the grouping of cliques attempted to include as many individuals as possible (Borgatti, Everett, & Freeman, 2002).

**Network centralization.** This indicator is measured by individual centrality. It is the extent to which an individual is central to a network and is determined by calculating individuals' degree, closeness, and betweeness scores. *Degree* refers to the number of direct links with others, *closeness* refers to whether an individual can reach others easily in the network, and *betweeness* refers to whether an individual can mediate between any other two individuals in the network. For the present study, degree was considered as the main criterion for individual centrality because individuals can directly reach one another (closeness) and do not need mediators (betweeness) in online settings.

*Analysis*

UCINET 6.96, a network analysis software program (Borgatti et al., 2002), was used to calculate the network indicators. Pajek (Nooy et al., 2005) was used to display graphical representations (communigrams) of the communication networks.

## Results

For RQ1 about network sizes, the multiplex network consisted of 344 individuals. Among the uniplex networks,


DOI: 10.1002/asi

TABLE 2. Network measures.

| Network | Size | No. of linkages | Density | No. of cliques |
|---|---|---|---|---|
| Informational | 254 | 446 | .007 | 1 |
| Network | 161 | 246 | .01 | 1 |
| Emotional | 122 | 170 | .01 | 0 |
| Thanks | 50 | 96 | .04 | 2 |
| Esteem | 37 | 40 | .03 | 0 |
| Whole | 344 | 624 | .005 | 12 |

TABLE 3. Threads.

| No. of posts | Frequency (%) | Year/Month[a] |
|---|---|---|
| 2 | 83 (49.4) | 04/02, 04/04, 04/07, 04/08, 05/04, 05/05, 05/06, 05/07, 05/08, 05/09, 05/10, 05/11, 05/12, 06/01, 06/03, 06/04, 06/05, 06/06 |
| 3 | 43 (25.6) | 05/04, 05/05, 05/07, 05/08, 05/09, 05/10, 05/12, 06/03, 06/04, 06/05, 06/06, 06/07 |
| 4 | 15 (8.9) | 04/06, 04/07, 05/04, 05/05, 05/06, 05/07, 05/08, 05/12, 06/02, 06/03, 06/06 |
| 5 | 8 (4.8) | 05/09, 06/01, 06/03, 06/04, 06/05, 06/07 |
| 6 | 2 (1.2) | 06/03 |
| 7 | 4 (2.4) | 05/04, 05/05, 06/05, 06/06 |
| 8 | 5 (3.0) | 05/09, 06/02, 06/03, 06/04, 06/07 |
| 9 | 2 (1.2) | 05/05 |
| 10 | 3 (1.8) | 05/08, 05/10, 06/03 |
| 11 | 1 (.60) | 06/03 |
| 12 | 2 (1.2) | 05/08 |
| Total | 168 (100.0) | |

[a]A thread appeared at this time. For example, a thread containing two posts appeared in February 2004, April 2004, July 2004, and so on.

the information network was the largest of 254 individuals, followed by network with 161, emotion with 122, thanks with 50, and esteem network with 37 (Table 2).

For RQ2 about thread size, the shortest thread consisted of two posts communicated by two individuals while the longest thread consists of 12 posts by six individuals. Overall, 75% of the threads consisted of three or fewer posts. The most frequent type of thread consisted of two posts by two individuals exchanging informational types of social support (49.4%, see Table 3). The dyadic interactions also appeared most frequently over time (18 months of 30-months sampling time frame, see Table 3). Among the 344 individuals who participated in a thread, the majority (71.5%) wrote one post, and 58 (16.9%) wrote two posts (see Figure 1).

For the number of links observed for the networks, the informational network had the highest number of links at 446, followed by network at 246, emotion at 170, thanks at 96, and esteem network at 40. No tangible types of support were identified during the observation period. Thus, for RQ3.1 relating to network density, the ratio of observed links over possible links ranged from .005 (the whole) to .04 (thanks) (see Table 2). The density of the whole and each of the five types of support networks also was displayed by communigrams from Figures 2 through 7. In each communigram, solid dots with numerals by the side indicate individuals participating in the PTT psychosis group while lines connecting dots indicate supportive communication between individuals. The thicker a line, the more communication links had been exchanged.

For associations between network density and support types, quadratic assignment procedures were used to measure correlations between the five networks, two networks at a time. All five networks correlated significantly with each other. Thus, for RQ3.2, no significant associations were found

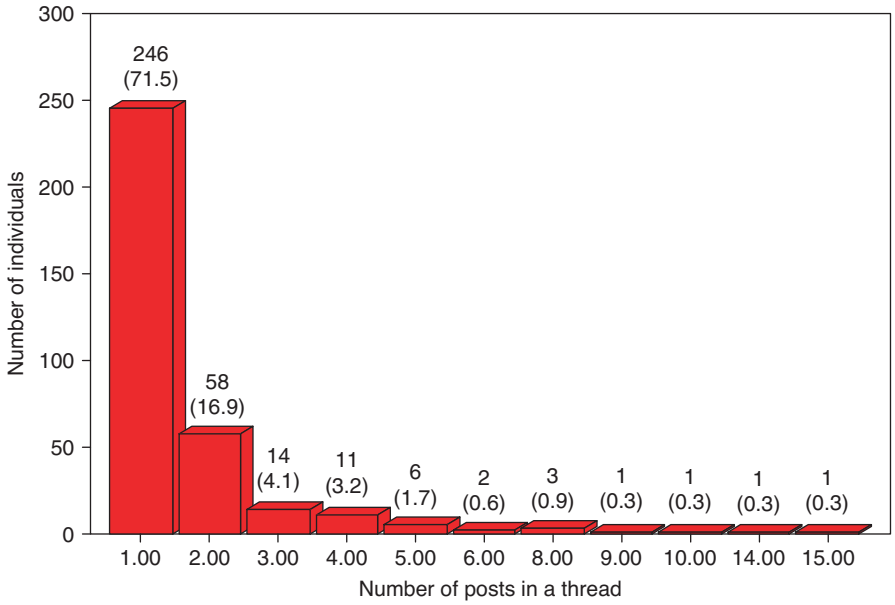



FIG. 1. Participation distribution of individuals in the threads.


DOI: 10.1002/asi

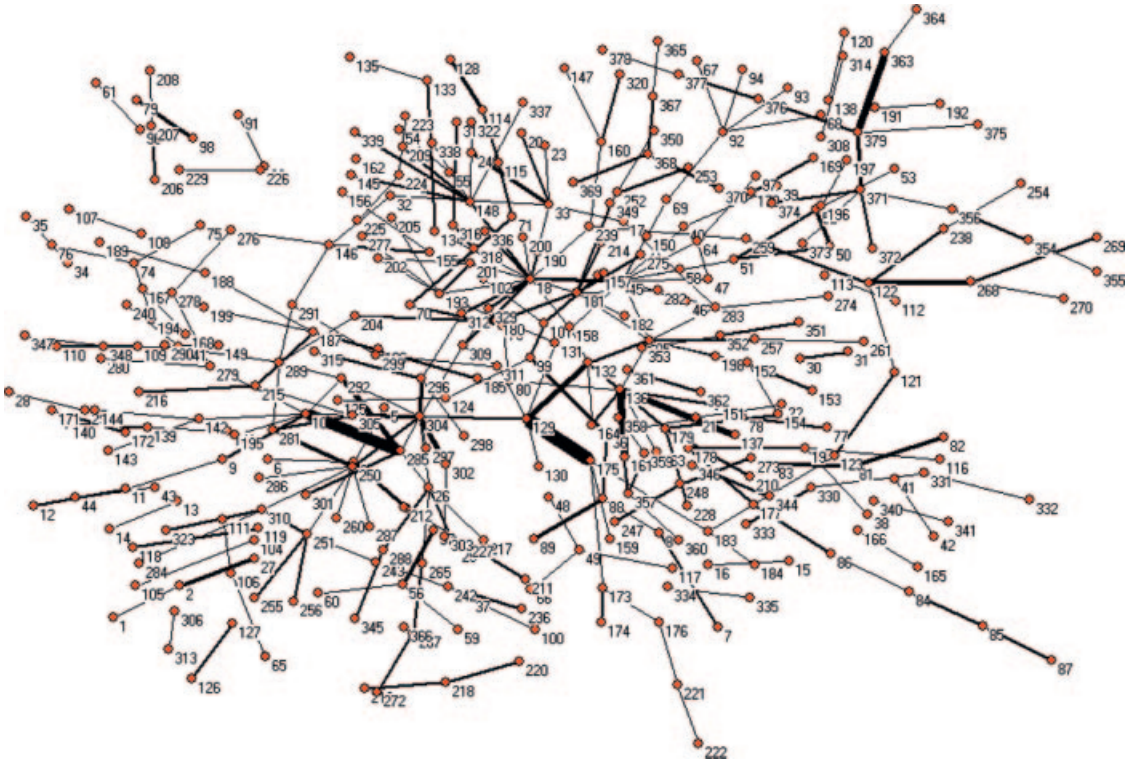


FIG. 2. Communigram of the multiplex PTT psychosis network.

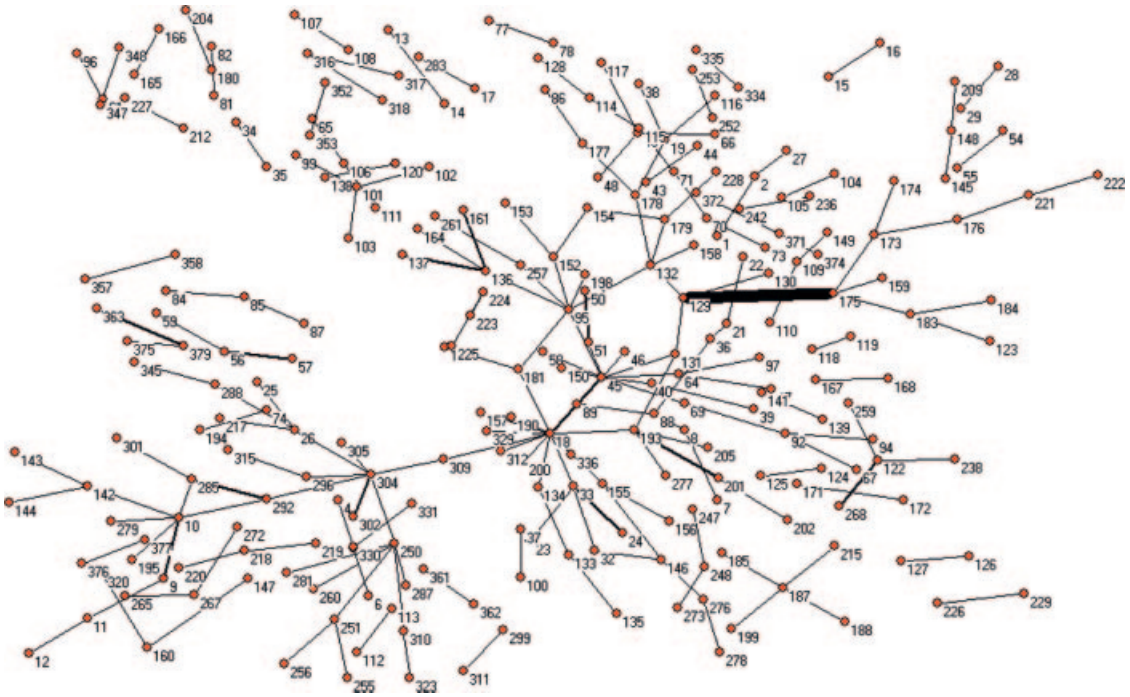


FIG. 3. Communigram of the informational communication network.

between network density and support types. Yet, the density values did suggest that the informational network had lower density (.007) than did the companionship-oriented networks such as esteem (.03), emotion (.01), and network (.01).

For RQ4 on clique formation, seven cliques formed within the whole network, two in the thanks network, and one in network (as in social support) network and one in information network. No cliques were formed in either emotion


DOI: 10.1002/asi

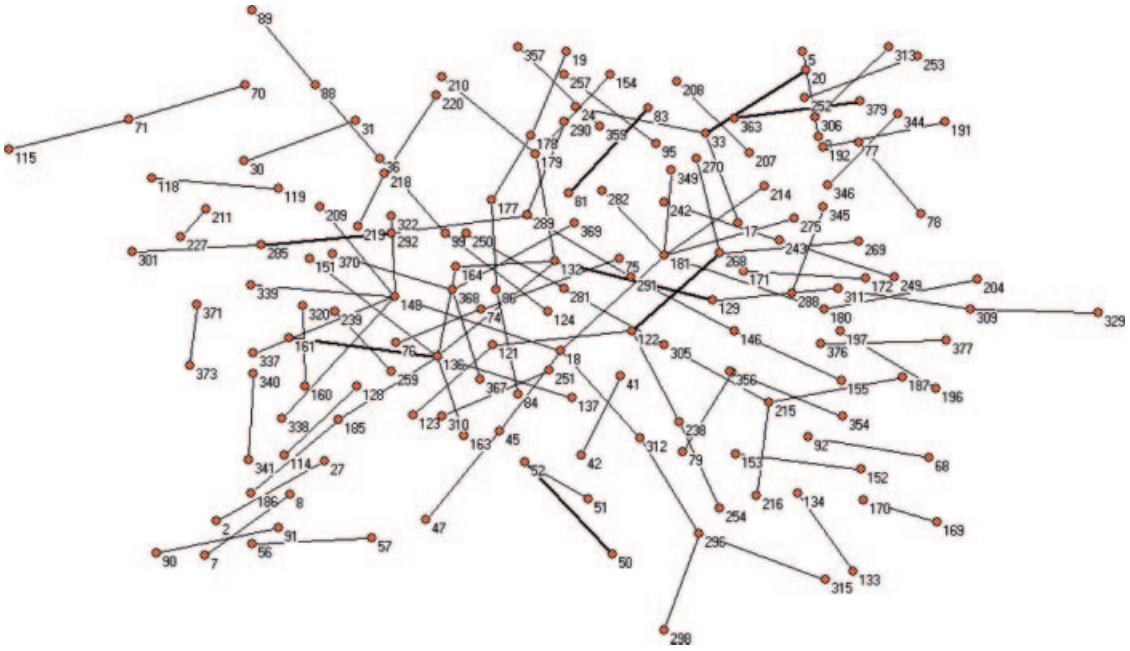


FIG. 4. Communigram of the network communication network.

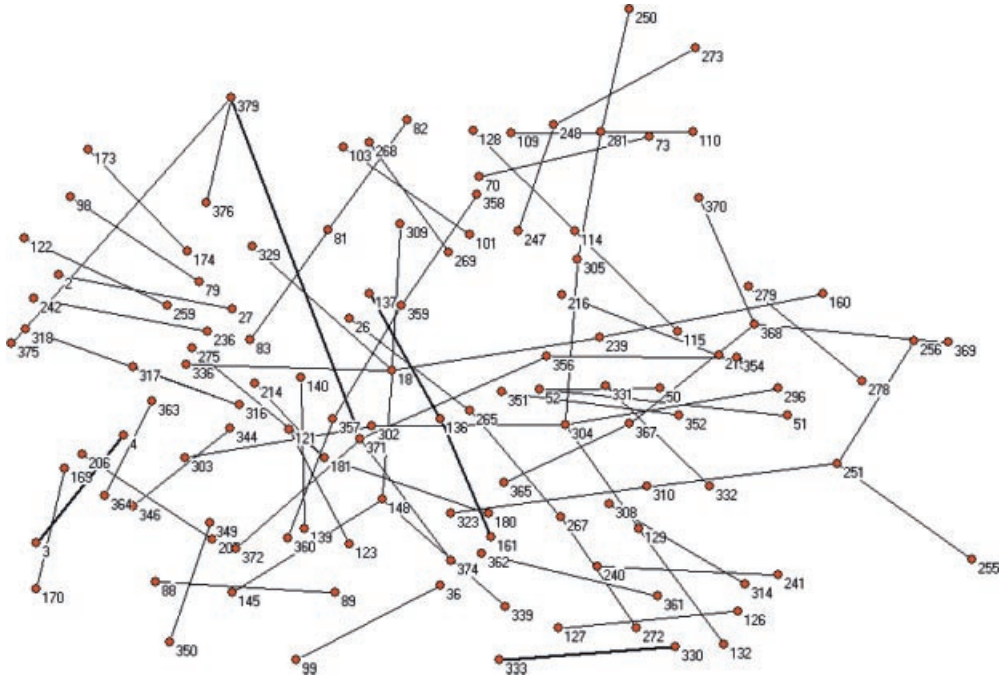


FIG. 5. Communigram of the emotional communication network.

or esteem networks (Table 4). The largest clique consisted of four individuals. A hierarchical clustering of overlapping memberships revealed by UCINET among the seven cliques demonstrated that Cliques 1 and 2 had shared members who tended to communicate information and emotion messages among themselves. Cliques 3 and 4 tended to communicate information and express thanks among themselves (as these individuals formed cliques in information and thanks networks as well). Cliques 5 and 6 shared information and network messages (as three of them formed cliques in network [as in social support] network). Clique 7, with only one individual overlapped with Clique 1, shared information and emotional content among themselves.

Finally, for RQ5, the network is relatively centralized. The overall individual centrality scores indicated that a group of seven central individuals dominated about one fifth of the total communication links within the PTT psychosis group. A closer examination of the outdegree (i.e., outgoing


DOI: 10.1002/asi

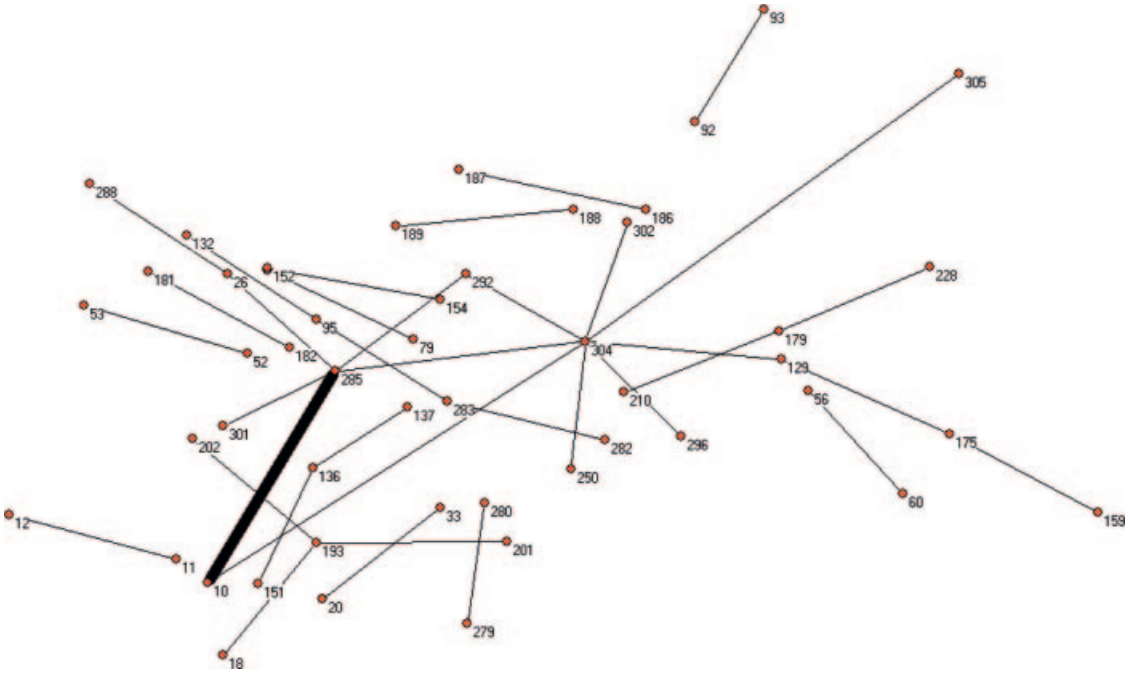



FIG. 6. Communigram of the thanks communication network.

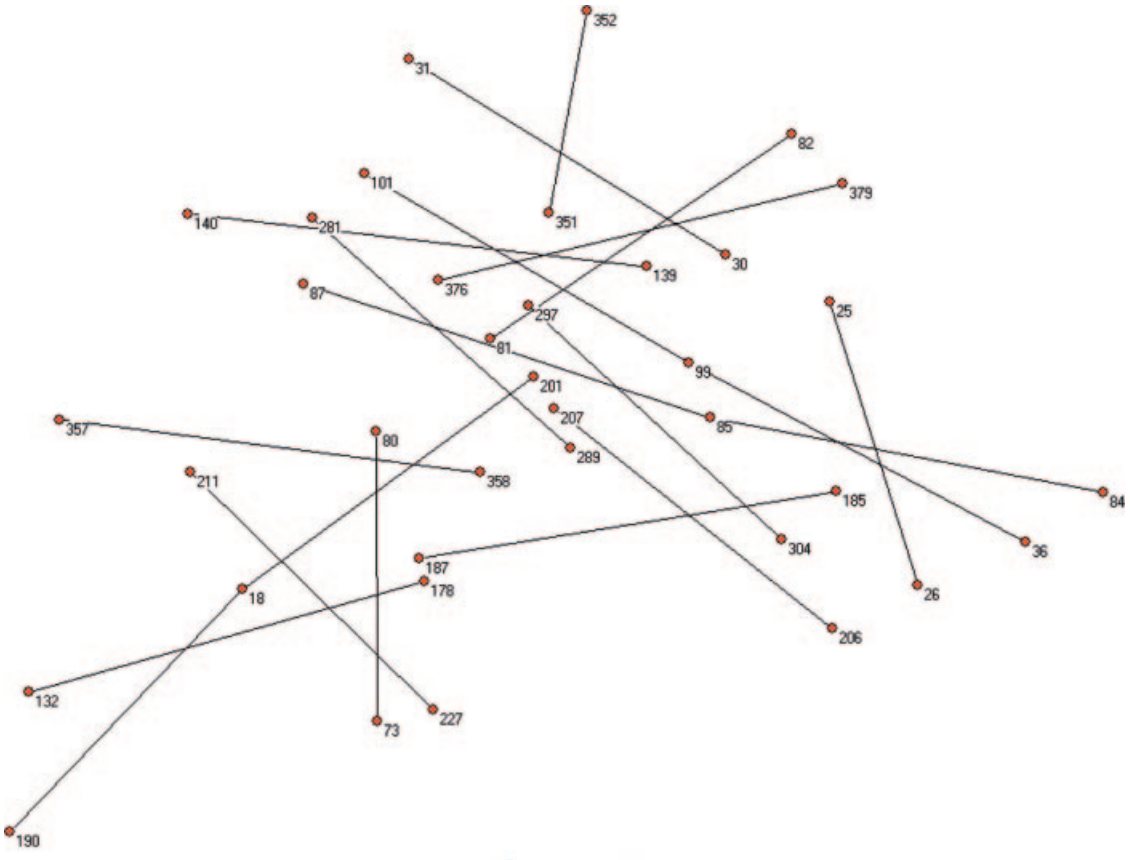



FIG. 7. Communigram of the esteem communication network.


DOI: 10.1002/asi

TABLE 4. Cliques.

| Whole | Informational | Network | Thanks |
|---|---|---|---|
| (1) **18**,[a] 193, 201 | 10, 285, 292 | 132, 136, 164 | 10, 285, 304 |
| (2) 193, 201, 202 | | | 285, 292, 304 |
| (3) *10*[b], **285**, 292, **304** | | | |
| (4) 26, **285**, **304** | | | |
| (5) 95, **132**, *136* | | | |
| (6) **132**, *136*, 164 | | | |
| (7) **18**, 309, 329 | | | |

[a]numbers in bold indicates central support providers.
[b]numbers in italics indicate central support receivers.

TABLE 5. Individual centrality: Support providers.

| Individual numbers | Outdegree (types of social support) | Indgree |
|---|---|---|
| 18 | 20 (informational, emotional) | 6 |
| 304 | 16 (informational, thanks) | 6 |
| 285 | 15 (thanks) | 7 |
| 379 | 12 (emotional, network, informational) | 1 |
| 132 | 12 (informational, network) | 2 |
| 129 | 11 (informational) | 10 |
| 181 | 11 (network, emotional) | 3 |

TABLE 6. Individual centrality: Support seekers.

| Individual numbers | Indgree (types of social support) | Outdegree |
|---|---|---|
| 136 | 19 (network, emotional) | 2 |
| 45 | 13 (informational) | 0 |
| 175 | 12 (informational) | 4 |
| 129 | 10 (informational, emotional) | 11 |
| 10 | 10 (thanks) | 8 |
| 148 | 9 (network, emotional) | 2 |
| 250 | 9 (informational) | 1 |

links) and indegree (i.e., incoming links) scores showed that individuals with the highest outdegree did not correspond to individuals with the highest indegree scores. That is, most of the people who provided supports (outdegree) were not the same people who received support (indegree). Only one person (No. 129) served as both support provider and receiver (Tables 5 and 6).

More central support providers were clique members than central support receivers (see Table 4, numbers in bold and in italics, respectively). An examination of central individuals' communication content showed that three types of support were provided and received within the PTT psychosis group: information (Members 304 and 129), emotion/network (Member 181), and a combination of information, emotion, and network (Members 18, 379, and 132). Thus, the communication links made by these central members reflected to some extent the overall communication pattern and content transmitted within the PTT psychosis group.

## Discussion

The goal of this article was to describe online supportive interactions within a bulletin board type psychosis support group using a network approach. Supportive communication content was analyzed using five types of social support derived from previous literature and a preliminary examination of the present data: information, emotion, network, esteem, tangible, and thanks. The results indicated that the most exchanged support types were information and network links, and no tangible support was communicated during the sampling period. All types of supportive communication networks were relatively sparse, yet small groups of cliques with different provisions of support types still formed within the psychosis group. Most of the online supportive interactions (76%, see Table 3) exchanged at dyadic and triadic levels: Two to three posts exchanged among two to three people. The multiplex network was highly centralized: About 1% of group members dominated almost 20% of the overall communication. In fact, the supportive provision and receiving communication were centralized among different group members. That is, most central members were either support providers or support receivers, but rarely performed both roles. In adition, central members each specialized in different types of support provision: information; emotion and network; and emotion, network, and information.

I will discuss the overall findings with implications for future studies under two areas: (a) social support in online settings and (b) weak-tie network and supportive interactions. Finally, limitations of the study will be discussed.

### *Social Support in Online Settings*

Within the PTT psychosis group, the most frequently occurring type of social support was informational, followed by network support. Individuals afflicted with psychological discomfort need suggestions, advice, and factual inputs as well as companionship and connections. These findings are consistent with previous studies on various kinds of health problems (e.g., Bambina, 2005; Braithwaite et al., 1999; Coulson, 2005). In these circumstances, informational support facilitates actions to solve problems while network support promotes comfort. In addition to the properties of each type of social support, the unique characteristics of the online setting may have had an impact upon what was provided to the support seekers.

The online environment provides opportunities to extend individuals' personal networks. As a result, the likelihood of greater expertise being provided by the personal networks also increases (Walther & Parks, 2002). Thus, participants from all over the world may offer advice and suggestions to support seekers based on their conditions. Consequently, each uniplex PTT psychosis network was lager than those reported in FtF support networks and the core discussion networks. This indicates that individuals are willing to discuss important matters with many strangers. This finding corresponds with the observation reported in the previous


DOI: 10.1002/asi

study where the density of nonfamily discussion networks has increased over the past two decades (McPherson et al., 2006). In addition, as the size of an individual's personal network increases, more social relations can be established between support seekers and similar others, which facilitates the use of network support (Braithwaite et al., 1999).

Moreover, most bulletin-type sites work via asynchronous written communication. This gives participants plenty of time to deliberate more thoughtfully before they reply to messages. As a result, according to Braithwaite et al. (1999), "better or more thoughtful advice" (p. 129), informational support, is prevalent in online support groups.

Furthermore, the results of the current study showed that no tangible aid was communicated among the psychosis group during the observation period. This finding is reasonable because no actual physical assistance, such as exchange of money and goods, can be performed within online settings (Walther & Boyd, 2002). However, the other type of supportive communication content identified in the study is expression of thanks, which often accompanied the request for informational aids. Within the psychosis group, more than 50% of thanks links centralized around three individuals (Figure 6; Members 285, 10, and 304). These three individuals also formed cliques among themselves (Table 4). Thus, it can be suggested that the expression of thanks, a gesture of gratitude and politeness, is an important element in sustaining online supportive interaction within a weak-tie network. Even more so, as mentioned earlier, this study was conducted in Taiwan, a country characterized by a collectivistic culture (Hofstede, 2001) and high-context communication (Hall, 1976). In high-context communication, people do not disclose personal information about themselves: "It appears that individualists would engage in more self-disclosure than would collectivists" (Gudykunst & Matsumoto, 1996, p. 35). Growing up in Taiwan, it is not easy to reveal information about oneself, especially regarding a sensitive topic such as psychosis. The mere action of posting on the psychosis board indicates that an individual may have this type of problem. Thus, the postings of thanks may indicate tacit understanding within this cultural context.

Other findings concern support seekers and providers in online settings. First, more than two thirds of participating individuals within the PTT psychosis group wrote one post (Figure 1). For those who participated in more than one type of support network, some became central members who dominated the information flow in the PTT psychosis network. These central members (i.e., stars) traversed various types of support networks. In other words, multiplex relations facilitated social support, which also has been demonstrated in previous research (Johnson, 1993). Second, central members found in the PTT psychosis group fell into two categories: support providers and support receivers. It seems that some people were always providing support and that some were always on the receiving side. The possible explanation for the observed pattern is that these central members, especially the providers, expressed certain types of authority in the posts, which is important in creating group identity (Galegher et al., 1998). Thus, they showed a preference for providing certain types of social support. The results indicated that three types of experts emerged during the observation period: information; emotion/network; and a combination of information, emotion, and network. Most of them directly expressed their expertise. For example, of the two information experts, one identified himself or herself as a psychology student and gave a lot of medication advice while the other showed his or her knowledge of psychosis disease by discussing all kinds of professional treatment and medication. In addition, some of the emotion and network experts established their authority by sharing their own experiences.

### *Weak-Tie Networks and Supportive Interactions*

The full sequence of supportive interactions observed in traditional FtF relations, which consists of support activation, support provision, support receipt, and accompanying reactions and responses between a target and a helper, was not observed within the PTT psychosis group. This has two implications. First, the results indicate that 50% of supportive interactions were exchanged between two people and consisted of informational content. In other words, only the first two phases of the full sequence of supportive interactions were realized in an online setting. Most of the group members, either targets or helpers, posted once in a thread during the sampling period. Participation over time was rather transitory. Individuals just came in, wrote posts, waited for replies, and left. This may indicate a brief and functional approach to obtaining online social support. Second, for threads consisting of more than two posts, a third or fourth party usually joined and ended the discussion. After a target activated a thread, more than one helper provided support. Thus, it appears that supportive messages spread through the PTT psychosis network by "walks," a kind of trajectory in which the sequences of links are unrestricted. In an online supportive setting, it is an influential process in which targets' and helpers' messages "change in each other's beliefs or attitudes" through interaction (Borgatti, 2005, p. 58).

In terms of network measures, although online social support is exchanged in relatively large networks (Walther & Boyd, 2002), few studies have provided such detailed network descriptors for an online support network as does the present study. The results show that the PTT psychosis support network is characterized by low density, high centralization, and small cliques. No significant associations between network density and types of social support were found. It is possible that the observed pattern is an artifact of the uniformly low densities of the supportive networks. However, it also is possible that the network characteristics of the PTT psychosis group are representative of all other weak-tie networks dealing with psychosis problems. Over time, this network is loosely connected among strangers, with most participating infrequently. Stars and cliques were indeed formed within the PTT network, and most of the clique members provided the same type of social support. From a reciprocal point of view, individuals are more likely to engage in


DOI: 10.1002/asi

interaction when they are exchanging resources and sharing common attributes (Monge & Contractor, 2003). Within the PTT network, those who exchanged informational, network, and emotional support and shared their common affliction of personal discomfort stuck together. However, timing also is important. A closer examination of the data showed that clique members often appeared in the same or consecutive months during the sampling period. Only three (Members 10, 18, and 95) appeared on the psychosis board over time.[1] Thus, it is possible that strangers were cliqued together because they happened to seek and provide social support at the same time. Clique membership, like the overall participation pattern, is rather transitory as well.

Still, more studies need to be conducted to examine the overall pattern of distribution between support providers and seekers, and to explore the explanation behind the pattern.

Finally, as mentioned earlier, it is not easy for the Taiwanese to disclose personal information about themselves. Although anonymity is provided within online settings, it takes more than courage to change that cultural characteristic. Consequently, the PTT psychosis support group may even be a "weaker-tie" network due to the cultural influence. Future study should consider culture as a background variable, and cross-cultural studies should be conducted.

### *Limitations*

A major limitation of the present study is its external validity. As discussed earlier, the results of the study may be specific to a certain group in a particular cultural context only. All findings may be suggestive and serve as a reference point for future studies in the area of online supportive interaction.

The other limitation is more technical in nature. The clique detection provided by UCINET can calculate only binary data rather than valued data on which the current study is based. This limits the chance to display the strength of communication and connection within each clique.

## Acknowledgment

The author thanks the anonymous reviewers for their valuable suggestions.

## References

Albrecht, T.L., & Adelman, M.B. (1987). Communicating social support: A theoretical perspective. In T.L Albrecht & M.B. Adelmen (Eds.), Communicating social support (pp. 18–39). Newbury Park, CA: Sage.

Albrecht, T.L., Burleson, B.R., & Goldsmith, D. (1994). Supportive communication. In M.L. Knapp & G.R. Miller (Eds.), Handbook of interpersonal communication (2nd ed., pp. 419–449). Thousand Oaks, CA: Sage.

Bambina, A.D. (2005). Understanding online social support: The interplay of Internet technology, social networks and social support. Unpublished doctoral dissertation, Columbia University.

[1]Member 10 participated in June 2004, September 2005, February 2006, and March 2006. Member 18 participated in May 2005, August 2005, September 2005, December 2005, April 2006, and May 2006. Member 95 participated in May 2005, August 2005, September 2005, and March 2006.

Bauer, M. (2000). Classical content analysis: A review. In M. Bauer & G. Gaskell (Eds.), Qualitative researching with text, image and sound (pp. 131–151). Thousand Oaks, CA: Sage.

Bertero, C.M. (2000). Types and sources of social support for people afflicted with cancer. Nursing and Health Sciences, 2, 93–101.

Bjornsdottir, G. (1999). Online social support for individuals concerned with heart disease: Observing gender differences. Proceedings of the AMIA Symposium (pp. 681–685).

Borgatti, S.P. (2005). Centrality and network flow. Social Networks, 27, 55–71.

Borgatti, S.P., Everett, M.G., & Freeman, L.C. (2002). UCINET for Windows: Software for Social Network Analysis. Harvard, MA: Analytic Technologies.

Braithwaite, P.O., Waldron, V.R., & Finn, J. (1999). Communication of social support in computer-mediated groups for people with disabilities. Health Communication, 11(2), 123–151.

Burleson, B.R., Liu, M., Liu, Y., & Mortenson, S.T. (2006). Chinese evaluations of emotional support skills, goals, and behaviors: An assessment of gender-related similarities and differences. Communication Research, 33(1), 38–63.

Burleson, B.R., & MacGeorge, E.L. (2002). Supportive communication. In M.L. Knapp & J.A. Daly (Eds.), Handbook of interpersonal communication (2nd ed., pp. 374–424). Thousand Oaks, CA: Sage.

Burt, R.S. (1992). Structural holes: The social structure of competition. Cambridge, MA: Harvard University Press.

Caplan, G. (1974). Support systems and community mental health. New York: Behavioral Publications.

Cobb, S. (1976). Social support as a moderator of life stress. Psychosomatic Medicine, 38, 300–314.

Cotton, S.R., & Gupta, S.S. (2004). Characteristics of online and offline health information seekers and factors that discriminate between them. Social Science & Medicine, 59, 1795–1806.

Coulson, N.S. (2005). Receiving social support online: An analysis of a computer-mediated support group for individuals living with irritable bowel syndrome. CyberPsychology & Behavior, 8(6), 580–584.

Cutrona, C.E., & Russell, D.W. (1990). Types of social support and specific stress: Toward a theory of optimal mating. In B.R. Sarason, I.G. Sarason, & G.R. Pierce (Eds.), Social support: An interactional view (pp. 319–366). New York: Wiley.

Cutrona, C.E., & Suhr, J.A. (1992). Controllability of stressful events and satisfaction with spouse support behaviors. Communication Research, 19(2), 154–174.

Durkheim, E. (1897). The Emile Durkheim Archive. Last retrieved December 24, 2005, from http://durkheim.itgo.com/

Eichhorn, K.C. (2008). Soliciting and providing social support over the Internet: An investigation of online eating disorder support groups. Journal of Computer-Mediated Communication, 14, 67–78.

Franzen, E. (2000). Does the Internet make us lonely? European Sociological Review, 16(4), 427–438.

Galegher, J., Sproull, L., & Kiesler, S. (1998). Legitimacy, authority, and community in electronic support groups. Written Communication, 15, 493–530.

Gudykunst, W.B., & Matsumoto, Y. (1996). Cross-cultural variability of communication in personal relationships. In W.B. Gudykunst, S. Ting-Toomy, & T. Nishida (Eds.), Communication in personal relationships across cultures (pp. 19–56). Thousand Oaks, CA: Sage.

Hall, E.T. (1976). Beyond culture. New York: Doubleday.

Hanneman, B.A., & Riddle, M. (2005). UCINET Tutorial. Retrieved December 2006, from http://faculty.ucr.edu/~hanneman/nettext/

Haythornthwaite, C. (2000). Online personal networks: Size, composition and media use among distance learners. New Media & Society, 2, 195–226.

Herring, S.C. (2004). Content analysis for new media: Rethinking the paradigm. In New research for new media: Innovative Research Methodologies Symposium Working Papers and Readings (pp. 47–66). Minneapolis: University of Minnesota School of Journalism and Mass Communication.


DOI: 10.1002/asi

Hiller, H.H., & Franz, T.M. (2004). New ties, old ties, and lost ties: The use of the Internet in Diaspora. New Media & Society, 6(6), 731–752.

Himelboim, I. (2008). Reply distribution in online discussion: A comparative network analysis of political and health newsgroups. Journal of Computer-Mediated Communication, 14, 156–177.

Hllebec, V., Manfreda, K.L., & Vehovar, V. (2006). The social support networks of Internet users. New Media & Society, 8(1), 9–32.

Hofstede, G. (2001). Culture's consequences: International differences in work-related values. Newbury Park, CA: Sage.

Hosti, O. (1969). Content analysis for the social sciences and humanities. Reading, MA: Addison-Wesley.

Johnson, J.D. (1993). Organizational communication structure. Norwood, NJ: Ablex.

Johnson, J.D. (1996). Information seeking: An organizational dilemma. Westport, CT: Quorum Books.

Jones, S. (1999). Doing Internet research: Critical issues and methods for examining the Net. Thousand Oaks, CA/London: Sage.

Marsden, P.V. (1987). Core discussion networks of Americans. American Sociological Review, 52, 122–131.

McPherson, M., Smith-Lovin, L., & Brashears, M.E. (2006). Social isolation in America: Changes in core discussion networks over two decades. American Sociological Review, 71, 353–375.

Monge, P.R., & Contractor, N.S. (2003). Theories of communication networks. New York: Oxford University Press.

Moss, G.E. (1973). Illness, immunity, and social interaction. New York: Wiley.

Nooy, W.D., Mrvar, A., & Batagelj, V. (2005). Exploratory social network analysis with Pajek. New York: Cambridge University Press.

Robinson, J.D., & Turner, J. (2003). Impersonal, interpersonal, and hyperpersonal social support: Cancer and older adults. Health Communication, 15(2), 227–234.

Tardy, R.W., & Hale, C.L. (1998). Getting "plugged in:" A network analysis of health-information seeking among "stay-at-home moms." Communication Monographs, 65, 336–357.

Tichon, J.G., & Shapiro, M. (2003). The process of sharing social support in cyberspace. CyberPsychology & Behavior, 6(2), 161–170.

Walther, J.B., & Boyd, S. (2002). Attraction to computer-mediated social support. In C.A. Lin & D. Atkin (Eds.), Communication technology and society: Audience adoption and uses (pp. 153–188). Cresskill, NJ: Hampton Press.

Walther, J.B., & Parks, M.R. (2002). Cues filtered out, cues filtered in: Computer-mediated communication and relationships. In M.L. Knapp & G.R. Miller (Eds.), Handbook of interpersonal communication (2nd ed., pp. 529–563). Thousand Oaks, CA: Sage.

Wasserman, S., & Faust, K. (1994). Social network analysis: Methods and applications. New York: Cambridge University Press.

Wellman, B. (1992). Which type of ties and networks give what kinds of social support? In E. Lawler, B. Markovsky, C. Ridgewau, & H. Walker (Eds.), Advances in group processes (Vol. 9, pp. 207–235). Greenwich, CT: JAI Press.

Wellman, B. (2001). Computer networks as social networks. Science, 293(5537), 2031–2034.

Wright, K. (2000). Computer-mediated social support, older adults, and coping. Journal of Communication, 50(3), 100–118.

Xie, B. (2008). Multimodal computer-mediated communication and social support among older Chinese Internet users. Journal of Computer-Mediated Communication, 13, 728–750.

Yoo, E-Y., & Robbins, L.S. (2008). Understanding middle-aged women's health information seeking on the Web: A theoretical approach. Journal of the American Society for Information Science and Technology, 59(4), 577–590.


DOI: 10.1002/asi